Data Science Capstone — Literature Review

Project: MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

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1. Affective Computing Fundamentals

Picard's foundational work [1] is where affective computing as a discipline really starts. Her central claim was that computers could be built to recognize and respond to emotional states, not just ignore them as noise. That idea sounds almost obvious now, but at the time it wasn't, and it's still the premise MoodBand rests on: that internal affective states leave measurable traces a system can learn to read.

Decades later, Conceição et al. [2] show what that premise looks like in practice with modern tools. They build on Russell's circumplex model, which maps emotion along two axes, valence and arousal, and use a deep-seeded clustering algorithm to classify emotional states from signals collected by a wearable wristband, testing it across three separate benchmark datasets. What makes this paper useful for MoodBand specifically is that it's working with the same kind of hardware form factor (a wrist-worn device) and the same underlying signal types we'd expect from a Fitbit, and it still manages to map those signals onto discrete emotional categories reasonably well.

Implication for MoodBand: together, [1] and [2] justify the project's core premise — that physiological signals can be mapped onto discrete emotional categories — and specifically justify grounding MoodBand's mood labels (happy, sad, stressed, angry, neutral) in the valence/arousal circumplex model rather than inventing an ad hoc labelling scheme. This directly shaped the mood categories used throughout the system design (Sections 6–8).

2. Wearable Physiological Sensing

Moving from theory to the specific question of wearables, Saganowski et al.'s systematic review [3] is a useful starting point. They find that heart rate, HRV, and electrodermal activity dominate the literature as the go-to signals for everyday emotion recognition, and they flag how much harder things get once you move out of the lab and into real, everyday conditions. That's exactly the setting MoodBand is targeting.

On the consumer-hardware side specifically, [4] is probably the single most reassuring paper on this list. Using the LifeSnaps dataset, built from actual Fitbit Sense smartwatches rather than research-grade sensors, the authors predict anxiety from everyday wearable data and get results suggesting a plain consumer device carries useful signal, not just noise.

That said, [5] is a good reality check on the same point. The authors ran into some genuinely annoying constraints when pulling Fitbit data for ML purposes: step counts reading zero while the device is just charging, HR data missing when the device isn't worn properly, all requiring manual filtering. Most notably, Fitbit only computes HRV while the user is asleep and relatively still, which would otherwise mean HRV could not be relied upon as a live, daytime signal.

Implication for MoodBand: papers [4] and [5] both use Fitbit devices specifically, which is directly relevant since Fitbit is also the device selected for this project (see Section 7.2 / Section 5.7). The choice of Fitbit was driven primarily by a requirement outside the scope of this literature review on its own: the companion app must work for both Android and iOS users. Fitbit's cloud-based API serves data identically regardless of the user's phone platform, whereas device-native alternatives such as Apple's HealthKit only work on iOS. This means the HRV constraint identified in [5] — that Fitbit only computes HRV overnight — carries over directly into MoodBand's design, and the classification pipeline (Section 7.3) is built around that constraint rather than around it being designed away.

3. Mood/Emotion Classification Models

Once the signals are available, the question is what model actually does the classifying. The systematic review in [6] gives a pretty clear answer: classical machine learning methods, Random Forest and Support Vector Machines in particular, show up in the large majority of wearable-based stress detection studies, with Random Forest especially holding up well on EDA and HRV features.

[7] adds two things on top of that. First, the authors design their classification approach specifically for resource-constrained, on-device deployment — directly relevant given MoodBand's classifier must run on a mobile phone rather than a server. Second, they compare personalized models against generalized ones and find personalized models can win out even with limited data per person, which matters given MoodBand's own user trial will necessarily be small.

Implication for MoodBand: [6] directly influenced the choice of classifier family — Random Forest is used as the primary candidate classifier (Section 7.3) specifically because it is the best-performing classical model on EDA/HRV-style wearable features in the reviewed literature, rather than defaulting to a heavier deep-learning approach that would be harder to justify running on a phone. [7] influenced two further decisions: favouring lightweight, resource-constrained model architectures suitable for on-device or near-device inference (Section 9), and adopting a personalized-model evaluation strategy alongside a generalized one, given the project's necessarily small participant sample (Section 7.6).

4. Accessible Feedback Design

The last piece of MoodBand, getting feedback to the user through non-visual channels, is backed by a smaller but still useful set of papers. The iFeeling system [8] is about as close a precedent as exists: it takes a detected emotional state and renders it as a distinct vibration pattern on a phone, built explicitly with visually impaired users in mind.

[9] widens the lens a bit. It's a recent, large-scale systematic literature review spanning two decades (2004 to 2024) of haptic assistive tools for blind and low-vision users, drawing on 132 papers, and it's useful less for any single finding and more as a way to check iFeeling's design choices against the broader literature.

Finally, [10] grounds all of this in actual user experience. Yeo and Chew interviewed visually impaired people in Singapore about how they use mobile apps day to day, and the picture isn't entirely rosy: screen readers often don't work as reliably as one would hope. Their recommendations explicitly call for combining audio and haptic feedback rather than relying on either alone, and for taking users' emotional wellbeing seriously as a design goal.

Implication for MoodBand: [8] directly shaped the haptic feedback channel — MoodBand's vibration-based mood cues (Section 7.4, Section 8) are modelled on iFeeling's approach of rendering a detected emotional state as a distinct phone-vibration pattern. [9] broadened this into a fuller design vocabulary: MoodBand's use of graduated or directional vibration patterns (e.g., descending vibration intensity for anger, see Table 1) draws on patterns documented across this wider body of research rather than the single iFeeling precedent alone. [10] most directly shaped a specific design rule: because screen readers were found to behave unreliably in practice, MoodBand pairs every visual cue with a corresponding audio or haptic equivalent by default (Section 7.4), rather than treating a screen-reader-compatible UI as sufficient on its own.

5. Synthesis

Put together, the pieces MoodBand depends on each have reasonable support behind them. Physiological signals can be mapped onto discrete moods [1], [2]. A consumer wearable like Fitbit is a workable, if imperfect, way to collect those signals [3]–[5]. Classical, lightweight models are a sensible fit for the classification step given the deployment constraints [6], [7]. And haptic feedback in particular has real precedent for communicating affective state to visually impaired users [8]–[10]. What doesn't exist yet, as far as this review found, is a single system that ties all four pieces together into one mobile-first pipeline — a gap this project aims to address.

6. Sensor Recommendations Across Reviewed Literature

Table 2 maps each reviewed paper to the physiological sensors or signals it used or recommended, alongside the sensors MoodBand will actually move forward with.

Paper	Sensors Used / Recommended
[1] Picard (1997)	N/A — conceptual/foundational text, no specific sensor
[2] Conceição et al.	Heart rate / blood-volume pulse (PPG), electrodermal activity (EDA), skin temperature (via Empatica E4 wristband)
[3] Saganowski et al. (review)	Heart rate, HRV, electrodermal activity (identified as the dominant signals across the wearable-sensing literature)
[4] Velu et al. (LifeSnaps/Fitbit)	Heart rate, Fitbit-derived stress score, sleep, step count (Fitbit Sense)
[5] King et al.	Heart rate, HRV (sleep-only), step count / motion (Fitbit intraday data)
[6] Pataca et al. (review)	Electrodermal activity, HRV (found Random Forest performs best specifically on these two)
[7] Irfan et al.	ECG (minimal wearable sensing)
[8] iFeeling	N/A — haptic output/rendering paper, not a sensing paper
[9] Jiang et al. (review)	N/A — haptic feedback review, not sensing
[10] Yeo & Chew	N/A — qualitative UX study, not sensing
Sensors MoodBand will use	**Heart rate (continuous, real-time) and motion/accelerometer (continuous), with heart-rate variability (HRV) incorporated as a supplementary overnight-only signal**

Table 2: Sensors recommended per paper vs. sensors selected for MoodBand

These are the signals consistently available through a consumer-grade, cross-platform-friendly wearable API (Fitbit, via the Google Health API). Electrodermal activity and skin temperature — used in the lab-grade Empatica E4 wristband in [2] and recommended generally in [3] and [6] — are not exposed by any mainstream consumer wearable's public API and are therefore out of reach for this project. ECG-based sensing as explored in [7] likewise requires dedicated ECG-capable hardware (e.g., a chest strap, or Apple Watch's ECG app) not available through Fitbit's consumer API. MoodBand's sensor selection therefore reflects the intersection of what the literature recommends and what a consumer-grade, cross-platform wearable API actually exposes, with HRV demoted to a supplementary overnight signal per the constraint identified in [5].

7. Device Selection Summary

To select a specific wearable, four candidate devices — Fitbit, Apple Watch, Samsung Galaxy Watch, and WHOOP — were compared against price, cross-platform support, developer API accessibility, real-time data availability, and development-environment fit. The full comparison is documented separately (MoodBand — Wearable Device Selection: Comparative Analysis, submitted for supervisor review); this section summarizes that analysis and its conclusion.

One piece of context shaped the timing of this decision: the legacy Fitbit Web API (the one referenced by [4] and [5]) was retired by Google — developer support ended September 30, 2026, with full shutdown October 30, 2026 — and replaced by the new Google Health API. This is a net positive for the project: the legacy API required a justification request to unlock sub-daily heart-rate data, whereas the new API provides 5-second-resolution heart rate by default with no special approval step.

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Price (new / used)	$70–130 / $30–80	$220–250 / $120–180	$170–310 / $100–150	Bundled with membership
Works on Android?	Yes	No — iPhone-only, structurally	Yes	Yes
Works on iOS?	Yes	Yes	No — continuous streaming is Android-only	Yes
API approval needed?	No (new Google Health API)	No (HealthKit)	Only for Samsung's cloud API (1–2 wks)	Free API, but returns no data without paid membership
Real-time daytime HR?	Yes — 5-sec resolution	Yes (workout mode)	Yes — on-device Sensor SDK	No — no continuous/raw HR via API
Matches lit. review?	Yes — [4], [5] used Fitbit directly	No direct precedent	No direct precedent	No direct precedent

Table 3: Condensed device comparison (full analysis in the separate device-selection document)

Decision: Fitbit, integrated via the Google Health API, was finalized as the project's sensing device. The deciding factor was the hard requirement that the companion app support both Android and iOS: this structurally rules out Apple Watch (pairs with iPhone only) and Samsung Galaxy Watch (continuous real-time streaming is Android-only via the Samsung Health Sensor SDK), regardless of their other technical merits. Fitbit's cloud-based API is the only option on the shortlist that serves identical data regardless of phone platform. WHOOP was ruled out independently of the platform question, since its API exposes no continuous or raw heart-rate data under any circumstance and requires an active paid membership just to return data at all.

References

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